

30TAI 智能应用开发指导手册

1. 开发流程及概念

1.1 开发流程

1.2 重要概念

2. 开发过程

2.1 基础环境

- 上位机：Win10 系统
- 开发板：FMQL30TAI • 悟净开发板
- 板载系统环境：Ubuntu20.04
- 开发板启动方式：SD 卡启动
- Icraft 版本：v3.33.1
- modelzoo 网址：

https://www.modelscope.cn/collections/icraft_modelzoo-18b52923d4854f

- tutorial 教程网址：

<https://gitee.com/mxh-spiger/tutorial-runtime/tree/tt3.33.1/>

2.2 Win 系统安装 Icraft（支持 Win7-Win11）

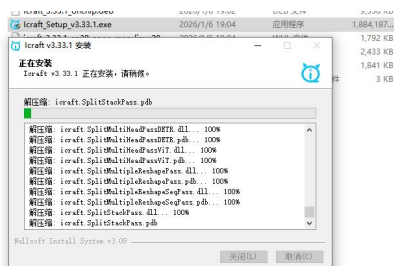
- 推荐先阅读 “复旦微 AI_Icraft 工具更新（v3.33.1）发布通知-FMQL30TAI.pdf”，了解

Icraft 相关功能点

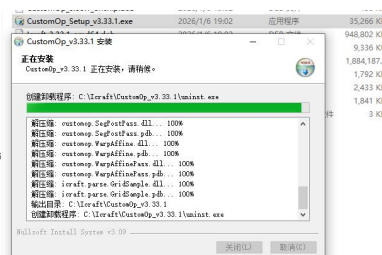
- 下载地址：

download.fdwzhb.com/data/04_FPAI/02_Icraft/v3.33.1/Icraft安装包/			
Index of /data/04_FPAI/02_Icraft/v3.33.1/Icraft安装包			
Name	Last modified	Size	Description
Parent Directory	-	-	-
CustomOp_3.33.1_amd64.deb	2026-01-06 19:02	3.8M	
CustomOp_3.33.1_onchip.deb	2026-01-06 19:02	452K	
CustomOp_Setup_v3.33.1.exe	2026-01-06 19:02	34M	
Icraft_3.33.1_amd64.deb	2026-01-06 19:02	927M	
Icraft_3.33.1_onchip.deb	2026-01-06 19:02	9.1M	
Icraft_Setup_v3.33.1.exe	2026-01-06 19:04	1.8G	
Icraft-3.33.1-cp38-none-manylinux2014_aarch64.whl	2026-01-06 19:04	1.7M	
Icraft-3.33.1-cp38-none-manylinux2014_x86_64.whl	2026-01-06 19:04	2.4M	
Icraft-3.33.1-cp38-none-win_amd64.whl	2026-01-06 19:04	1.8M	
readme.md	2026-01-06 19:04	2.5K	

- 安装 Icraft_Setup_v3.33.1.exe (必须项)



- 安装 CustomOp_Setup_v3.33.1.exe (必须项)



- 安装成功后, Icraft 路径: C:\Icraft

此电脑 > 本地磁盘 (C:) > Icraft

名称	修改日期	类型	大小
 CustomOp_v3.33.1	2026/1/12 14:23	文件夹	
 CLI v3.33.1	2026/1/12 14:23	文件夹	
 CLI	2026/1/12 14:23	文件夹	

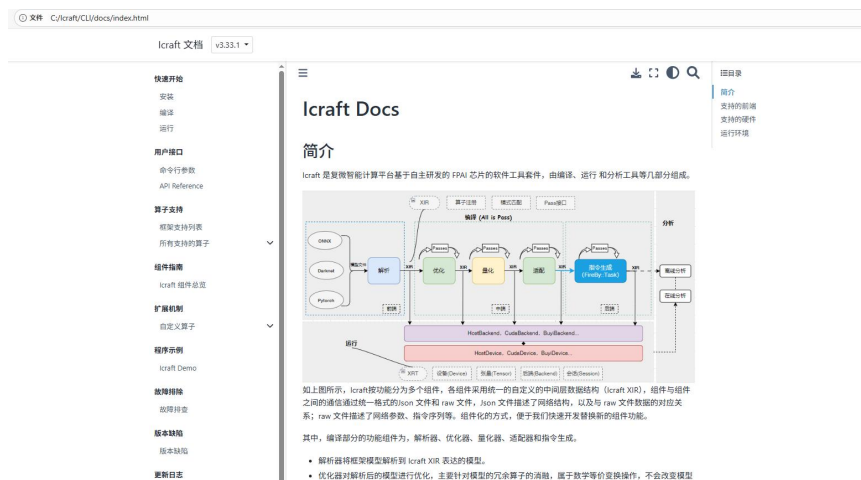
- [查看 Icraft 版本](#)

```
sunyuejun@sunyuejun C: > Icraft v3.33.1 > Icraft安装包
> icraft --version
Icraft 版本:
* 3.33.1

CLI 版本:
3.33.1.0-bccd576(2601061152)
sunyuejun@sunyuejun C: > Icraft v3.33.1 > Icraft安装包
```

- [查看 Icraft 文档](#)

sunyuejun@sunyuejun C: > Icraft v3.33.1 > Icraft安装包
icraft docs



• 安装 Icraft python 包（推荐项）

(1) 要求 python 3.8 环境

(2) 查看 python 环境是否已经安装其他版本 Icraft

```
sunyuejun@sunyuejun C:\> Icraft v3.33.1 Icraft安装包
> pip show icraft
WARNING: Ignoring invalid distribution -illow (c:\users\sunyuejun\appdata\local\program
te-packages)
Name: icraft
Version: 3.31.0
Summary: a whl for icraft
Home-page: UNKNOWN
Author: TangXiao
Author-email: tangxiao@fmsh.com.cn
License: GPL Licence
Location: c:\users\sunyuejun\appdata\local\programs\python\python38\lib\site-packages
Requires:
Required-by:
```

(3) 卸载其他版本

```
> pip uninstall icraft
Found existing installation: icraft 3.31.0
Uninstalling icraft-3.31.0:
  Would remove:
    c:\users\sunyuejun\appdata\local\programs\python\python38\lib\site-packa
    c:\users\sunyuejun\appdata\local\programs\python\python38\lib\site-packa
Proceed (Y/n)? y
Successfully uninstalled icraft-3.31.0
```

(4) 安装当前版本 Icraft

```
> pip install icraft-3.33.1-cp38-none-win_amd64.whl
WARNING: Ignoring invalid distribution -illow (c:\users\sunyuejun\appdata\local\program
te-packages)
Looking in indexes: https://pypi.tuna.tsinghua.edu.cn/simple
Processing c:\icraft v3.33.1\icraft安装包\icraft-3.33.1-cp38-none-win_amd64.whl
WARNING: Ignoring invalid distribution -illow (c:\users\sunyuejun\appdata\local\program
te-packages)
Installing collected packages: icraft
Successfully installed icraft-3.33.1
```

2.3 交叉编译环境搭建（适用于板载系统开发，不适用于裸机开发）

• 板载系统为 Ubuntu20.04-aarch64，并且板载资源有限，直接在板载系统编译可执行文件会很慢，还有可能由于内存问题编译不成功，所以需要在上位机 Win 系统构建交叉编译环境，

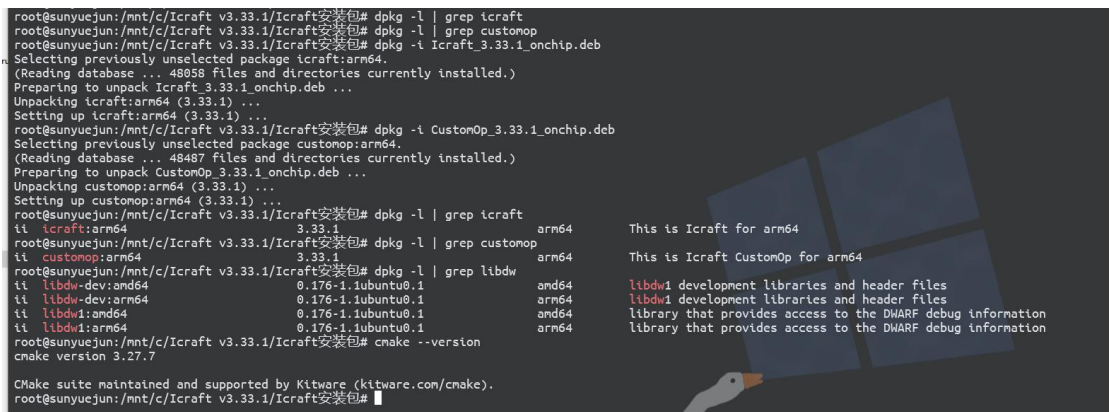
编译板载应用程序，生成可直接在板子上运行的可执行程序。

- 交叉编译环境的搭建方式有以下三种，根据开发习惯任意选择一种即可：

(1) 交叉编译虚拟机，根据教程在环境中安装对应版本的 Icraft，下载地址：



- 交叉编译环境安装 Icraft 方法



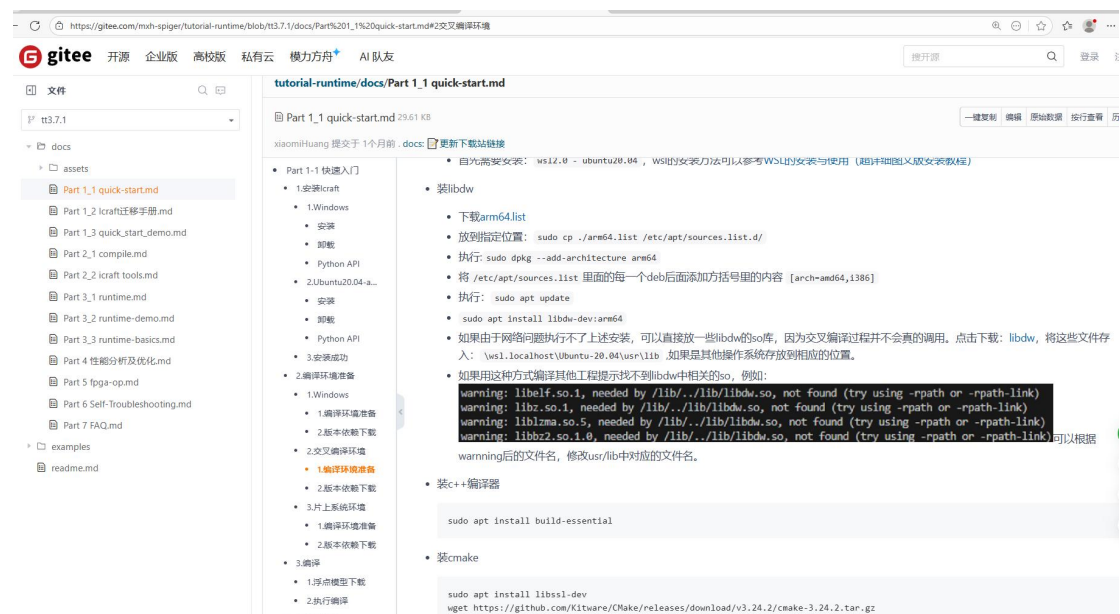
(2) Docker 环境，参考 tutorial 中 env_setup 部分进行安装使用



- (3) WSL，安装过程可参考以下网址：

https://blog.csdn.net/weixin_57367513/article/details/135001273

ubuntu20.04 系统中需要安装其他三方依赖库请参考：

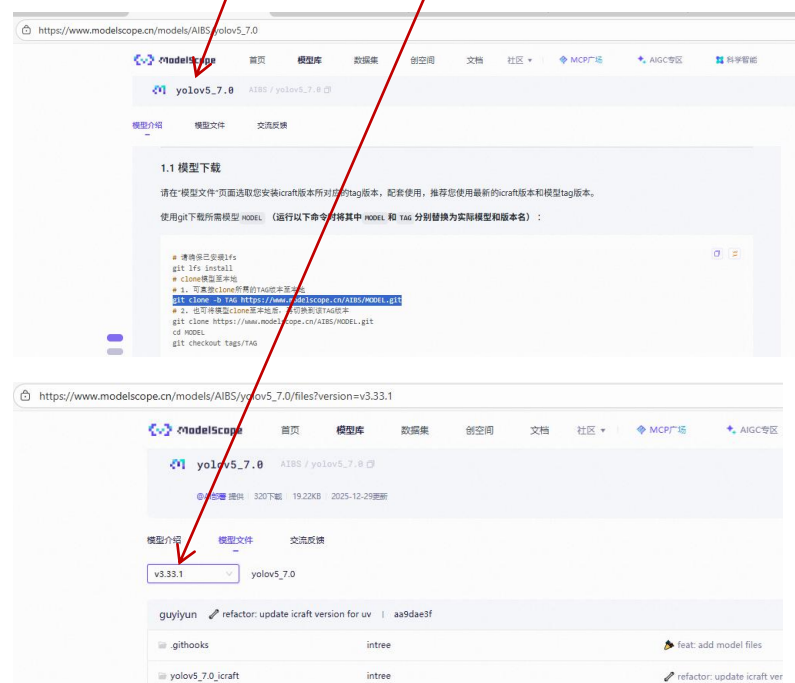


2.4 运行 modelzoo_yolov5_psin 示例工程准备工作

- modelzoo_yolov5_psin 工程下载地址:

https://www.modelscope.cn/models/AIBS/yolov5_7.0

- 下载方式: MODEL 为模型名, TAG 为 Icraft 版本



- 下载过程, 依赖 git 环境:


```
sunyuejun@sunyuejun C: > Icraft v3.33.1 > modelzoo
• > git --version
git version 2.42.0.windows.1
sunyuejun@sunyuejun C: > Icraft v3.33.1 > modelzoo
• > git lfs install
Git LFS initialized.
sunyuejun@sunyuejun C: > Icraft v3.33.1 > modelzoo
○ > git clone -b v3.33.1 https://www.modelscope.cn/AIBS/yolov5_7.0.git
Cloning into 'yolov5_7.0'...
remote: Enumerating objects: 695, done.
remote: Counting objects: 100% (103/103), done.
remote: Compressing objects: 100% (67/67), done.
remote: Total 695 (delta 61), reused 55 (delta 36), pack-reused 592
Receiving objects: 100% (695/695), 1.19 MiB | 358.00 KiB/s, done.
Resolving deltas: 100% (321/321), done.
Note: switching to 'aa9dae3feec08e3d006aa620888f8414797436df'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:
```

- 参考 readme 进行部署开发，下载的文件夹包含了所有必需文件，可直接进行编译验证



(1) 导出 pytorch 框架模型为 Torchscript/ONNX 格式用于 Icraft 编译输入

对于 python 环境管理工具 uv，不是必选项，如果本地有 anaconda 环境可自行构建 python 环境。

```
• ②_save_infer.py：用修改后保存的模型做前向推理，验证保存的模型与原模型是否一致

进入 yolov5_7.0_icraft/1_scripts 后，即可执行上述脚本：

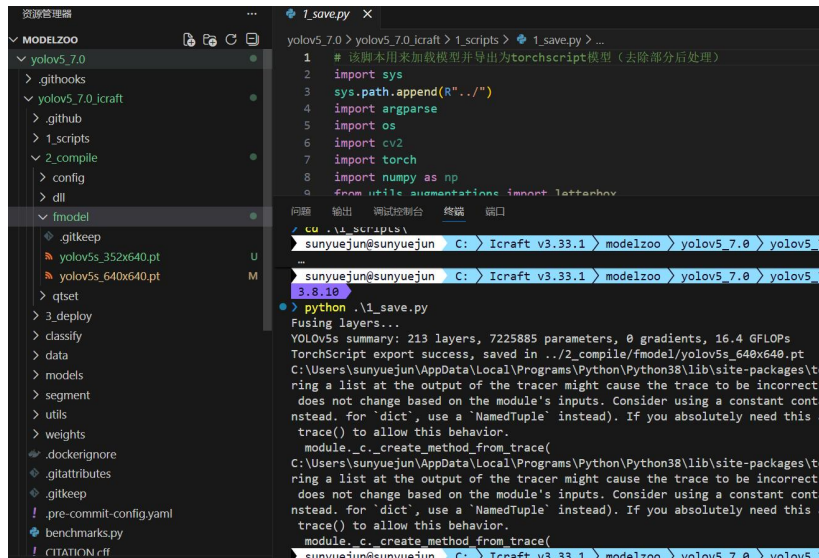
• 如果已安装uv，可跳过此步，否则请先使用以下命令安装（更多uv的安装和基础使用说明可参考 tutorial 中对应部署后端的 part ④ env_setup.md/3.3 Python环境管理工具uv的安装和使用(optional) 章节：

# 推荐使用官网源码安装
powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1 | iex"
# 也可pip安装
pip install uv

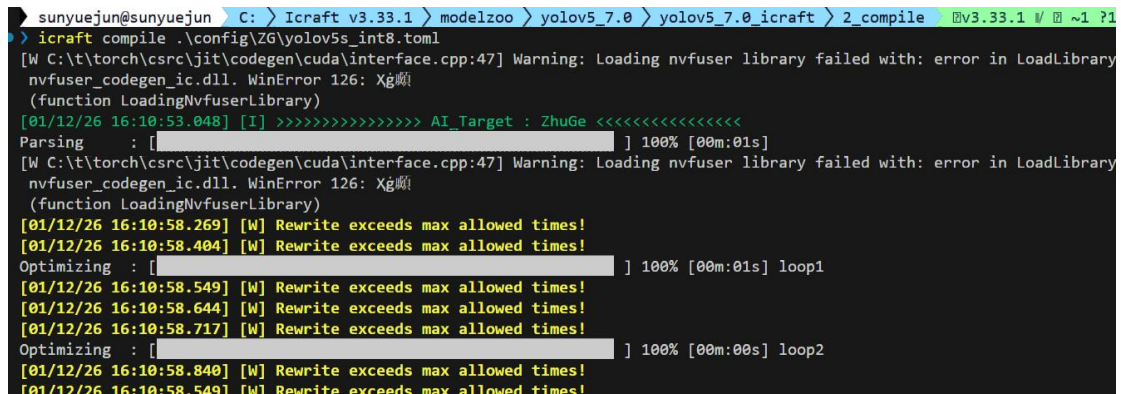
• 运行脚本：（首次在项目目录下使用uv命令运行脚本时，会自动加载配置项目环境，用时可能稍长）

# 在此模型目录
cd yolov5_7.0_icraft/1_scripts
uv run 1_save.py
```

以下为基于本地 anaconda 环境直接运行：python 1_save.py



(2) 编译模型，注意 30TAI 对应 NPU 架构为“诸葛/ZG/zg/zhuge”



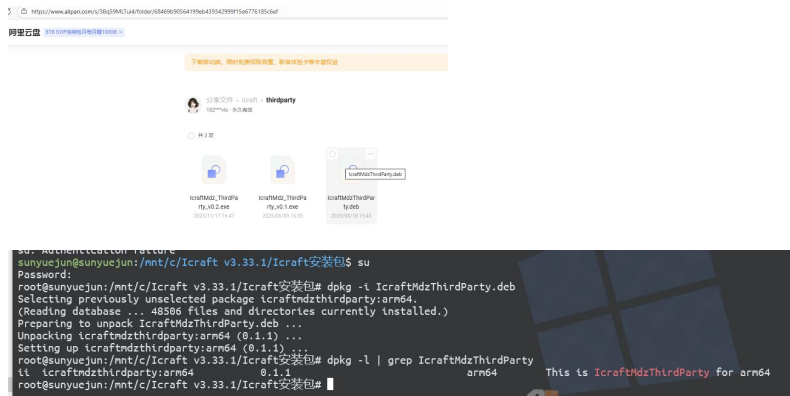
(3) 交叉编译板载运行 PSIN 工程

- 注意需要在交叉编译环境和板载系统中安装第三方依赖库（opencv 等）

IcraftMdzThirdParty.deb

下载地址：

<https://www.alipan.com/s/3Bq59MLTui4/folder/68469b90564199eb439342999f15e6776185c6ef>



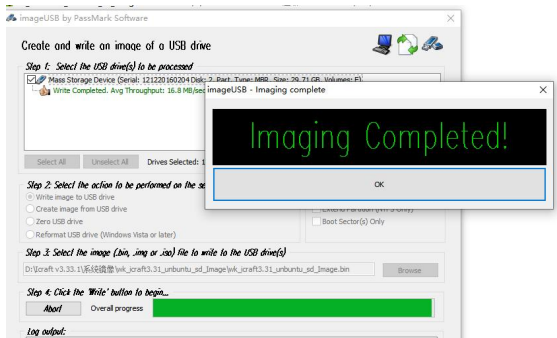
```
sunyuejun@sunyuejun:/mnt/c/Icraft v3.33.1/modelzoo/yolov5_7.0/yolov5_7.0_icraft/3_deploy/build_arm$ cmake -DLINUX_AARCH64=ON ..
-- The CXX compiler identification is GNU 9.4.0
-- The C compiler identification is GNU 9.4.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Found OpenCV: /ModelzooDeps/aarch64/Static (found version "4.5.5")
-- Configuring done
-- Generating done
-- Build files have been written to: /mnt/c/Icraft v3.33.1/modelzoo/yolov5_7.0/yolov5_7.0_icraft/3_deploy/build_arm
sunyuejun@sunyuejun:/mnt/c/Icraft v3.33.1/modelzoo/yolov5_7.0/yolov5_7.0_icraft/3_deploy/build_arm$ make -j
[ 50%] Building CXX object CMakeFiles/yolov5s_psin.dir/src/yolov5s_psin.cpp.o
```

2.5 制作 SD 卡镜像

- (1) 准备一张容量不少于 32GB 的 SD 卡
- (2) 板载系统镜像下载地址：



- (3) 按照烧写说明将镜像文件烧写至 SD 卡
 - 解压 wk_icraft3.31_unbuntu_sd_Image.rar 得到 wk_icraft3.31_unbuntu_sd_Image.bin
 - 将 wk_icraft3.31_unbuntu_sd_Image.bin 通过 imageUSB.exe 烧写至 SD 卡

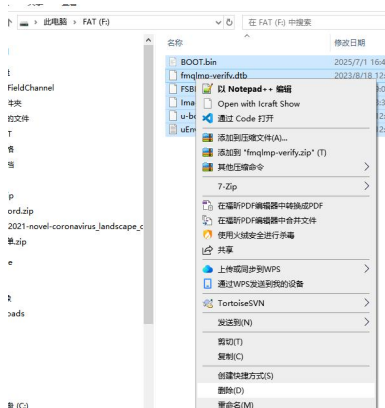


(4) 通过读卡器，读取 SD 卡，删除 FAT 分区的内容，将参考实现目录下的位流文件夹解压，拷贝至 SD 卡 FAT 分区

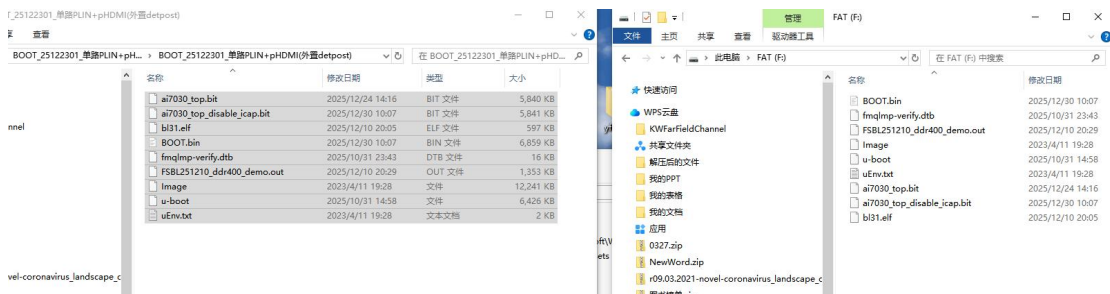
位流文件下载地址：根据需求选择不同的位流文件，如果是自研板卡需要基于 FPGA 工程文件修改管脚约束，生成适配自研板卡的位流文件。



• 删除 FAT 分区文件



• 将位流分件拷贝至 FAT 分区



2.6 启动悟净开发板（SD 卡启动）

（1）悟净开发板与外设等连接示意图



- (2) 通过串口/网口将悟净开发板与上位机连接，启动板卡
- (3) 通过串口登录板载系统，查看 IP，可根据需要修改 IP

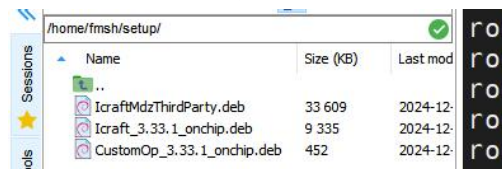
```
root@U:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.125.107 netmask 255.255.255.0 broad
```

2.7 板载系统安装 Icraft 和第三方依赖库 IcraftMdzThirdParty

- (1) 卸载系统自带 Icraft 环境

```
root@U:~# dpkg -r icraft
(Reading database ... 63821 files and directories currently installed.)
Removing icraft (3.31.0) ...
root@U:~# dpkg -r customop
(Reading database ... 63431 files and directories currently installed.)
Removing customop (3.31.0) ...
root@U:~# dpkg -l | grep icraft
root@U:~# dpkg -l | grep customop
root@U:~#
```

- (2) 将安装包上传至板载系统某个路径下



- (3) 安装 Icraft、CustomOp 和 IcraftMdzThirdParty

```
1 root@U:~# cd /home/fmsh/setup/
04 root@U:/home/fmsh/setup# dpkg -i Icraft_3.33.1_onchip.deb
04 Selecting previously unselected package icraft.
04 (Reading database ... 63415 files and directories currently installed.)
11 Preparing to unpack Icraft_3.33.1_onchip.deb ...
07 Unpacking icraft (3.33.1) ...
11 Setting up icraft (3.33.1) ...
07 root@U:/home/fmsh/setup# dpkg -i CustomOp_3.33.1_onchip.deb
07 Selecting previously unselected package customop.
07 (Reading database ... 63844 files and directories currently installed.)
11 Preparing to unpack CustomOp_3.33.1_onchip.deb ...
07 Unpacking customop (3.33.1) ...
11 Setting up customop (3.33.1) ...
07 root@U:/home/fmsh/setup# dpkg -i IcraftMdzThirdParty.deb
07 Selecting previously unselected package icraftmdzthirdparty.
07 (Reading database ... 63863 files and directories currently installed.)
11 Preparing to unpack IcraftMdzThirdParty.deb ...
07 Unpacking icraftmdzthirdparty (0.1.1) ...
11 Setting up icraftmdzthirdparty (0.1.1) ...
```

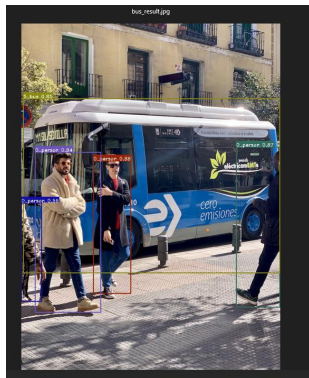
2.8 运行 modelzoo_yolov5 psin 示例工程

- (1) 将 2.4 章节交叉编译完成的 3_deploy 文件夹拷贝至板载系统某路径下
- (2) 运行程序

```
For details about running time message of the network, check the yolov5s_time.txt in path: ./logs/
root@U:/home/fmsh/3_deploy/build_arm# ./yolov5s_psin ../cfg/yolov5s_ZG_int8_demo.yaml
[2024-12-18 09:25:51.182] [info] Found model file at:../model/ZG/int8/yolov5s_ZG.json
[2024-12-18 09:25:52.082] [info] NetInfo init done
[2024-12-18 09:25:52.101] [info] Device initialization successful, version information:
[2024-12-18 09:25:52.101] [info]   icore: FMSHZGV3TECH-AID - 24160628
[2024-12-18 09:25:52.101] [info]   device: 25122301
[2024-12-18 09:25:52.101] [info]   Device Protocol: axi
[2024-12-18 09:25:52.101] [info]   Device Name: zg330atu
[2024-12-18 09:25:52.101] [info]   Device Parameters:
[2024-12-18 09:25:52.101] [info]     dma: 0x80000000
[2024-12-18 09:25:52.101] [info]     npu: 0x40000000
[12/18/24 09:25:52.338] [I] DetPostZG Version is 3.33.1.0
[12/18/24 09:25:52.338] [I] DetPostZG reg_base addr is : 0x100000800

Total_TotalTime: 18.94631 ms, Total_MemcpyTime : 2.558 ms, Total_HardTime : 8.973561 ms, Total_OtherTime :
s
Hardop_TotalTime: 13.65222 ms, Hardop_MemcpyTime: 0 ms,Hardop_HardTime : 7.724032 ms.
Customop_TotalTime: 0.65685 ms, Customop_HardTime : 0.13081 ms.
IO_TotalTime: 4.26987 ms, IO_MemcpyTime : 2.558 ms, IO_HardTime : 1.1187199 ms, IO_OtherTime : 0.59315 ms
*****
统计分析结果如下(The analysis results are as follows):
数据传入耗时(Data input time consumption):
Time(ms):2.558 Device:cdma
数据处理耗时(Data process time consumption):
Time(ms):1.71187 Device:npu
网络主体耗时(Network Backbone time-consuming):
Time(ms):13.6522 Device:npu
后处理硬算子耗时(Post CustomOp time-consuming):
Time(ms):0.65685 Device:DetPostZG
cpu算子耗时(CPU operator time consumption):
Time(ms):1.02422 Device:Null
*****
For details about running time message of the network, check the yolov5s_time.txt in path: ./logs/
root@U:/home/fmsh/3_deploy/build_arm#
```

(3) 结果保存于../io/output/文件夹



2.9 运行参考实现单路 PLIN+HDMI 工程

(1) 下载 runtimeapp 工程

download.fdwthb.com/data/04_FPAI/02_Icraft/v3.33.1/30TAI/参考实现/悟净开发板/单路PLIN+HDMI (外置detpost) 参考实现/25122301/runtimeapp/

Index of /data/04_FPAI/02_Icraft/v3.33.1/30TAI/参考实现/悟净开发板/单路PLIN+HDMI (外置detpost) 参考实现/25122301/runtimeapp

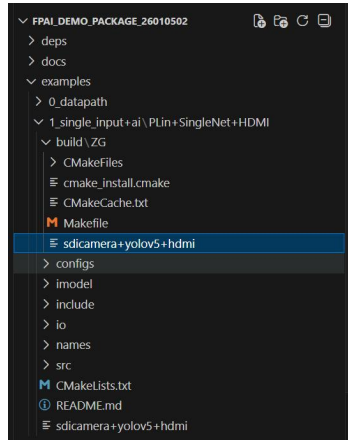
Name	Last modified	Size	Description
Parent Directory	-	-	-
fpai_demo_package_26010502.mds	2026-01-05 14:48	169	
fpai_demo_package_26010502.txt	2026-01-05 14:48	13K	
fpai_demo_package_26010502.zip	2026-01-07 17:24	312M	

(2) 按照 README.md 教程进行交叉编译

```
bash
sunyuejun@sunyuejun:/mnt/d/Icraft v3.33.1/fpai_demo_package_26010502/examples/1_single_input+ai/PLIN+SingleNet+HDMI$ cmake -S . -B build/Z
G -DTARGET_CHIP=ZG
-- The C compiler identification is GNU 9.4.0
-- The CXX compiler identification is GNU 9.4.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done

sunyuejun@sunyuejun:/mnt/d/Icraft v3.33.1/fpai_demo_package_26010502/examples/1_single_input+ai/PLIN+SingleNet+HDMI$ cmake --build build/ZG -j$(nproc)
[ 50%] Building CXX object CMakeFiles/sdicamera+yolov5+hdmi.dir/src/sdicamera+yolov5+hdmi.cpp.o
[100%] Linking CXX executable sdicamera+yolov5+hdmi
[100%] Built target sdicamera+yolov5+hdmi
sunyuejun@sunyuejun:/mnt/d/Icraft v3.33.1/fpai_demo_package_26010502/examples/1_single_input+ai/PLIN+SingleNet+HDMI$
```

(3) 将交叉编译生成的可执行文件 sdicamera+yolov5+hdmi 拷贝至 PLin+SingleNet+HDMI 目录下



(4) 将 fpai_demo_package_26010502 拷贝至板载系统某个路径下

(5) 运行程序

```
root@U:/home/fmsh/fpai_demo_package_26010502/examples/1_single_input+ai/PLin+SingleNet+HDMI# ./sdicamera+yolov5+hdmi ./
configs/ZG/sdicamera+yolov5+hdmi.yaml
[2024-12-18 09:41:44.100] [info] Log level set to 'debug'
[2024-12-18 09:41:44.101] [info] Loaded 80 class names from names/coco.names
[2024-12-18 09:41:44.111] [info] Device initialization successful, version information:
[2024-12-18 09:41:44.111] [info] icore: FMSHZGV3TECH-A1D - 24160628
[2024-12-18 09:41:44.111] [info] device: 25122301
[2024-12-18 09:41:44.111] [info] Device Protocol: axi
[2024-12-18 09:41:44.111] [info] Device Name: zg330atu
[2024-12-18 09:41:44.112] [info] Device Parameters:
[2024-12-18 09:41:44.112] [info] dma: 0x80000000
[2024-12-18 09:41:44.112] [info] npu: 0x40000000
[2024-12-18 09:41:44.114] [info] YOLOv5 Configuration:
[2024-12-18 09:41:44.114] [info] Name: yolov5s
[2024-12-18 09:41:44.114] [info] Confidence Threshold: 0.2
[2024-12-18 09:41:44.114] [info] IOU Threshold: 0.5
[2024-12-18 09:41:44.114] [info] Multi-label: false
[2024-12-18 09:41:44.114] [info] Number of Classes: 80
[2024-12-18 09:41:44.114] [info] FPGA NMS: false
[2024-12-18 09:41:44.114] [info] DETPOST: true
[2024-12-18 09:41:44.114] [info] Network Width: 640
[2024-12-18 09:41:44.114] [info] Network Height: 352
[2024-12-18 09:41:44.114] [info] Frame Width: 1920
[2024-12-18 09:41:44.114] [info] Frame Height: 1080
[2024-12-18 09:41:44.114] [info] Names Path: names/coco.names
[2024-12-18 09:41:44.114] [info] JSON Paths:
[2024-12-18 09:41:44.114] [info] - ./imodel/ZG/yolov5s_plin_352x640_ZG.json
[2024-12-18 09:41:44.114] [info] RAW Paths:
[2024-12-18 09:41:44.114] [info] - ./imodel/ZG/yolov5s_plin_352x640_ZG.raw
[2024-12-18 09:41:44.114] [info] Anchors:
[2024-12-18 09:41:44.114] [info] - [10.000000, 13.000000] [16.000000, 30.000000] [33.000000, 23.000000]
[2024-12-18 09:41:44.114] [info] - [30.000000, 61.000000] [62.000000, 45.000000] [59.000000, 119.000000]
[2024-12-18 09:41:44.114] [info] - [116.000000, 90.000000] [156.000000, 198.000000] [373.000000, 326.000000]
[2024-12-18 09:41:44.114] [info] YOLOv5 net0 json: ./imodel/ZG/yolov5s_plin_352x640_ZG.json, raw: ./imodel/ZG/yolov5s_p
lin_352x640_ZG.raw
DetPostZG_op_idx: 89, DetPostZG_op_id: 447
DetPostZG_out_idx: 89, DetPostZG_out_id: 447
```

(6) HDMI 输出显示示例

